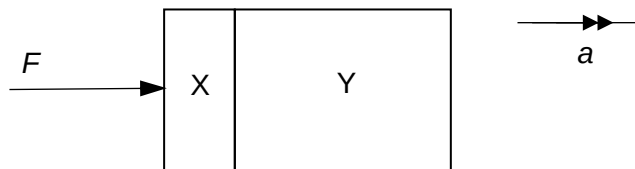


$$\therefore v = 7.75 / 0.200 = 38.8 \text{ m/s}$$

8

A

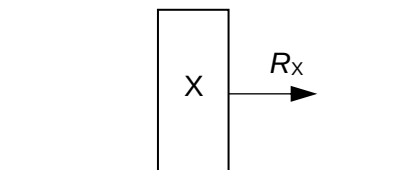
The resultant force acting on both blocks could be shown below.



$$\text{Hence, } F = 3ma$$

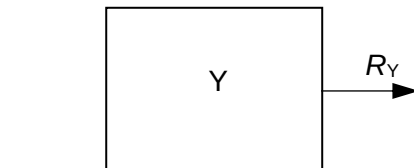
$$a = F / 3m$$

For Block X, let resultant force = R_X



$$R_X = ma = \frac{1}{3}F$$

For Block Y, let resultant force = R_Y



$$R_Y = 2ma = \frac{2}{3}F$$

3

Q

P

For an object to be in equilibrium with three coplanar forces acting on it